



Republic of the Philippines
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ITEP 414 - System Administration and Maintenance

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Part 1 – Company Profile

Pixelora

Company Overview

Company Name: Pixelora

Nature of Business: Software Development and Digital Solutions

Office Location: Santa Cruz, Laguna, Philippines

Pixelora is an emerging software development and digital solutions provider that builds dependable, modern technology for businesses and organizations. Its work spans web applications, mobile applications, business management systems, and cloud-based platforms designed to streamline day-to-day operations and support clients through digital transformation.

The company runs its operations from a single office floor and is made up of 20 employees spread across four departments: Information Technology, Human Resources, Finance, and Sales.

Vision

To be recognized as a dependable and forward-looking technology partner that empowers businesses to adapt, expand, and thrive through innovative digital solutions.

Mission

Pixelora is dedicated to building secure, dependable, and user-focused software that responds to genuine business challenges. By blending modern technology with continuous innovation and attentive service, the company helps its clients streamline operations and reach their goals.

Organizational Structure

Pixelora's organization is built around four core departments that report to company management:

Company Management

- Information Technology Department
- Human Resources Department
- Finance Department
- Sales Department

Information Technology Department

This department handles software development, upkeep of systems, technical support, network administration, and oversight of the company's overall IT infrastructure.

Human Resources Department

This department oversees recruitment, employee records, workplace policies, employee relations, and other personnel-related administrative matters.

Finance Department

This department is in charge of budgeting, payroll, accounting, financial recordkeeping, and the company's other financial transactions.

Sales Department

This department focuses on identifying prospective clients, nurturing customer relationships, presenting the company's service offerings, and pursuing new business opportunities.

Employee Distribution

Department	Employees	Primary Function
Information Technology	5	Software development, technical support, and IT infrastructure
Human Resources	4	Recruitment, employee management, and administration
Finance	5	Accounting, budgeting, payroll, and financial management
Sales	6	Client acquisition, customer relations, and sales

This employee breakdown forms the basis for planning the company's hardware, software, networking equipment, and other IT infrastructure requirements.

Part 2 – Enterprise Hardware Inventory

Hardware Inventory Overview

Pixelora needs dependable and scalable hardware to keep its daily operations running from software development and administrative work to financial processing, client communication, data storage, and network services.

The hardware quantities and types below were determined based on the needs of the company's 20 employees across its four departments.

Hardware Inventory

Asset ID	Hardware	Qty	Department / Location	Purpose
HW-001	Developer Desktop Computers	5	Information Technology	Software development, testing, virtualization, and technical administration
HW-002	Office Desktop Computers	9	Human Resources / Finance	Daily office work, accounting, documentation, payroll, and administrative tasks
HW-003	Business Laptops	6	Sales	Client presentations, communication, remote work, and sales activities
HW-004	Server (Rack/Tower)	1	Server / IT Area	Centralized file storage, user management, internal applications, and network services
HW-005	Business Router	1	Network Area	Routes traffic between the internal network and internet connection
HW-006	Managed Gigabit Switch	1	Network Area	Provides wired network connectivity for company devices
HW-007	Multifunction Network Printer	2	Shared Office Areas	Printing, scanning, and copying company documents

HW-008	UPS	3	Server / Network Area	Provides temporary backup power and protects critical equipment from power interruptions
HW-009	Wireless Access Point	2	Office Area	Provides reliable wireless connectivity throughout the office
HW-010	NAS Storage	1	Server / IT Area	Centralized storage and local backup of company files
HW-011	External Backup Drive	2	IT Department	Provides offline backup copies of important company data
HW-012	Monitors	25	All Departments	Provides displays for employee workstations, including additional monitors for developers

Hardware Selection and Justification

Desktop Computers

Pixelora will roll out 14 desktop computers across the Information Technology, Human Resources, and Finance departments.

Because software development, testing environments, and virtualization demand more processing power, the five IT staff members will be issued higher-spec developer workstations.

The remaining nine machines — used by the four Human Resources and five Finance employees — are standard business desktops suited for office applications, employee records management, accounting, payroll, documentation, and web-based systems.

Business Laptops

Six business laptops go to the Sales team, giving them the mobility they need for client meetings, presentations, communication, and work done outside the office.

Server

A single dedicated server delivers the company's centralized services, including file sharing, user management, internal applications, authentication, and other network-related functions.

Router

One business-grade router manages the flow of traffic between the company network and the internet connection, offering reliable routing and room for future expansion.

Managed Gigabit Switch

A single managed switch delivers wired connectivity to desktops, the server, printers, access points, and other network equipment. Because it is managed, it also gives the company finer control and the ability to organize the network logically as it grows.

Multifunction Network Printers

Two shared multifunction printers cover printing, scanning, and copying needs across the office, removing the need for individual printers per department.

Uninterruptible Power Supply (UPS)

Three UPS units safeguard critical infrastructure — the server, NAS, router, firewall, and switch — supplying short-term power during outages and cushioning equipment from sudden interruptions.

Wireless Access Points

Two access points blanket the single-floor office with Wi-Fi coverage, mainly serving laptops, mobile devices, and other approved wireless equipment.

NAS Storage

A dedicated NAS device centralizes storage for shared files and local backups, with access controlled by department and employee permissions.

External Backup Drives

Two external drives provide extra offline copies of critical data. These can be rotated so one copy always sits apart from the primary IT infrastructure.

Monitors

Twenty-five monitors will be distributed company-wide. Human Resources, Finance, and Sales staff each get a single display, while the five IT employees receive dual-monitor setups to boost productivity during development, testing, monitoring, and administration work.

Part 3 – Enterprise Software Inventory

Software Inventory Overview

To run its daily operations, Pixelora relies on a mix of operating systems, productivity suites, development tools, security software, remote-support utilities, and general-purpose programs. The software listed below was chosen to match the needs of each department and the company's broader IT environment.

Software Inventory

Software	Version	License	Purpose
Windows 11	25H2	Commercial Pro	Primary operating system for employee desktop computers and laptops
Ubuntu Server	24.04 LTS	Open Source	Server operating system for hosting internal services and applications
Microsoft 365	Current	Commercial Subscription	Provides Word, Excel, PowerPoint, Outlook, and other productivity applications
Visual Studio Code	Current	Free	Source code editor used by the IT Department for software development
Git	Current	Open Source	Provides version control for source code and development projects
GitHub Desktop	Current	Free	Provides a graphical interface for managing Git repositories
Oracle VirtualBox	Current	Open Source	Creates virtual machines for

			development, testing, and system administration
Google Chrome	Current	Free	Web browser for accessing web applications, online services, and testing websites
Microsoft Defender Antivirus	Built into Windows 11	Included with Windows	Provides malware and threat protection for Windows devices
AnyDesk	Current	Commercial / Business License	Provides remote access for authorized IT support and troubleshooting
7-Zip	Current	Open Source	Compresses and extracts files and archives

Software Selection and Justification

Windows 11 Pro

Windows 11 Pro is the standard operating system for desktops and laptops. Compared with consumer editions, the Pro edition adds management, security, and networking features better suited to a business setting.

Ubuntu Server

Pixelora's server infrastructure runs on Ubuntu Server, which offers a stable foundation for internal applications, file services, development environments, and other server-based functions the company needs.

Microsoft 365

Microsoft 365 gives employees access to core productivity tools — Word, Excel, PowerPoint, and Outlook — for document preparation, spreadsheets, presentations, email, and general business tasks.

Visual Studio Code

Mainly used by the IT Department, Visual Studio Code is a flexible code editor that developers use to write, modify, debug, and maintain software projects.

Git

Git lets the IT Department track version history, log changes, branch out development work, and collaborate without overwriting one another's code.

GitHub Desktop

GitHub Desktop puts a graphical front end on common Git tasks, helping developers manage repositories, branches, commits, and syncing alongside the command line.

Oracle VirtualBox

VirtualBox enables the IT team to spin up virtual machines on existing hardware for software testing, OS testing, training, and experimentation without touching the host workstation's main environment.

Google Chrome

Chrome serves as one of the company's main browsers, used by employees to reach web-based systems and online services, and by developers who rely on its built-in tools for testing and debugging web applications.

Microsoft Defender Antivirus

Defender Antivirus offers baseline endpoint protection against malware on Windows machines, working as one layer within the company's broader security setup and network protections.

AnyDesk

AnyDesk gives authorized IT staff remote access to employee computers, allowing many software issues to be diagnosed and fixed without a technician physically present at the workstation.

7-Zip

7-Zip handles compression and extraction of common archive formats, giving employees and IT personnel a lightweight way to manage compressed files, bundle files for transfer, and save on storage space.

Part 4 – Enterprise Network Inventory

Network Inventory Overview

To give its 20 employees internet access, secure communication, wired connectivity, and wireless access, Pixelora needs an organized and dependable network setup. The equipment below supports the company's four departments as well as its server and shared network resources.

Enterprise Network Inventory

Network Equipment	Qty	Location	Purpose
ISP Modem	1	Network / Server Area	Connects the company network to the Internet Service Provider
Business Router	1	Network / Server Area	Routes network traffic between the company's internal network and external networks
Hardware Firewall	1	Network / Server Area	Protects the internal network by filtering and controlling incoming and outgoing traffic
48-Port Managed Gigabit Switch	1	Network / Server Area	Provides centralized wired network connectivity for computers, servers, printers, access points, and other network devices
Wireless Access Point	2	Office Area	Provides wireless network coverage for laptops, mobile devices, and other authorized devices
48-Port CAT6 Patch Panel	1	Network / Server Area	Organizes and terminates Ethernet cable connections

			from different areas of the office
CAT6 Ethernet Cables	1 Box (305 m)	Entire Office	Provides wired Gigabit Ethernet connections between workstations, network devices, and the central network area
RJ45 Connectors	100 Pieces	IT / Network Area	Provides connectors for terminating CAT6 Ethernet cables used throughout the office

Part 5 – Enterprise Network Diagram

Pixelora Enterprise Network Topology

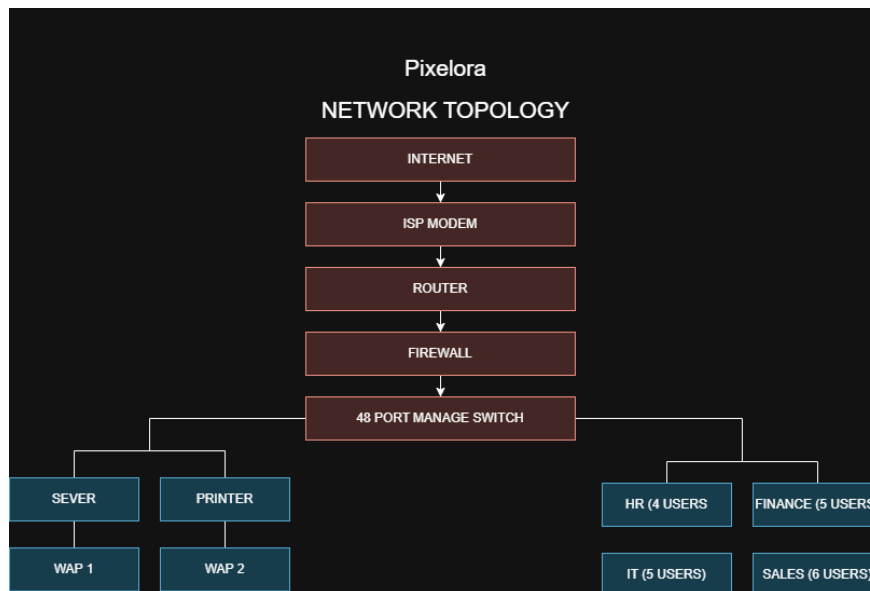


Figure 1. Pixelora Enterprise Network Topology

Pixelora's network is built around a central managed switch that ties together the company's departments and shared resources. Incoming internet traffic first passes through the ISP modem, then the router, then the firewall, before reaching the 48-port managed switch with the firewall acting as a buffer between the outside network and the company's internal one.

From that central switch, wired connections branch out to the server, the network printer, the wireless access points, and each of the four departments IT, HR, Finance, and Sales. The two access points handle Wi-Fi coverage inside the office, while the server delivers centralized services and resources to the rest of the network. Altogether, this layout gives Pixelora an organized, scalable network suited to its 20-person workforce.

Part 6 – System Administration Roles

Keeping an organization's technology running well takes a mix of IT professionals, each with a distinct set of duties, skills, tools, and areas of focus that together support the reliability, security, and uptime of its systems.

Helpdesk Technician

Responsibilities

A Helpdesk Technician is the front line of support for employees running into trouble with computers, software, accounts, printers, network access, and other IT resources. Their day-to-day includes diagnosing routine problems, walking users through fixes, installing or configuring software, resetting credentials, logging incidents, and passing along tougher cases to the right IT staff.

Skills

- Hardware and software troubleshooting
- Basic networking knowledge
- Windows administration
- Software installation and configuration
- User account management
- Problem-solving
- Technical documentation
- Communication and customer support

Common Tools

- Remote desktop and remote support software
- Ticketing and helpdesk systems
- Microsoft Windows administrative tools
- Microsoft 365 Admin Center
- Active Directory
- Command Prompt and PowerShell
- Hardware diagnostic utilities
- Network troubleshooting tools

Certifications

Certifications that support a career in helpdesk and IT support typically include:

- CompTIA A+
- CompTIA Network+

- Microsoft Fundamentals certifications
- ITIL Foundation

Network Administrator

Responsibilities

A Network Administrator keeps an organization's network infrastructure running, which means configuring and monitoring routers, switches, wireless access points, firewalls, and related equipment. They also resolve connectivity issues, manage IP addressing, watch network performance, roll out security measures, and keep communication between devices and systems dependable.

Skills

- TCP/IP networking
- IPv4 and IPv6 addressing
- Subnetting
- Routing and switching
- VLAN configuration
- Wireless networking
- Firewall configuration
- Network security
- Network monitoring and troubleshooting
- Basic network automation

Common Tools

- Cisco IOS
- Cisco Packet Tracer
- Wireshark
- PuTTY or other SSH clients
- PowerShell
- Command-line networking utilities such as ping, tracer, ipconfig, and nslookup
- Network monitoring systems
- Firewall management tools

Certifications

Certifications commonly held by Network Administrators include:

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+
- CompTIA Security+

- Cisco Certified Network Professional (CCNP) for more advanced networking roles

Cisco's current CCNA curriculum spans network fundamentals, network access, IP connectivity, IP services, security fundamentals, and automation and programmability.

Linux System Administrator

Responsibilities

A Linux System Administrator looks after servers and systems built on Linux. Their duties cover installing and configuring Linux servers, managing users and permissions, maintaining file systems, installing packages, setting up services, applying updates, monitoring performance, handling backups, and resolving server issues.

They also help lock down servers by managing access permissions, firewalls, authentication, and other security controls.

Skills

- Linux command-line administration
- User and group management
- File systems and permissions
- Package management
- Shell scripting
- Linux networking
- Service and process management
- System monitoring
- Backup and recovery
- Server security and troubleshooting

Common Tools

- Bash
- SSH
- systemd
- systemctl
- journalctl
- Vim or Nano
- APT and DNF package managers
- cron
- rsync
- firewalld

- Linux monitoring utilities such as top and htop

Certifications

Commonly held Linux certifications include:

- Red Hat Certified System Administrator (RHCSA)
- Red Hat Certified Engineer (RHCE)
- CompTIA Linux+
- Linux Foundation Certified System Administrator (LFCS)

The RHCSA credential confirms core system administration ability in Red Hat Enterprise Linux environments, covering system operation, storage, file systems, users, services, security, and basic container management.

Cloud Administrator

Responsibilities

A Cloud Administrator oversees an organization's cloud-hosted computing resources — deploying and maintaining virtual machines, configuring cloud storage, managing identities and permissions, setting up virtual networks, monitoring resources, running backups, applying security controls, and managing overall availability.

They also track resource usage and work to keep cloud infrastructure secure, reliable, and aligned with what the organization needs.

Skills

- Cloud computing concepts
- Virtual machine administration
- Cloud storage
- Virtual networking
- Identity and access management
- Security and governance
- Backup and disaster recovery
- Resource monitoring
- Operating system administration
- Basic automation and scripting

Common Tools

- Microsoft Azure
- Amazon Web Services (AWS)
- Google Cloud
- Microsoft Entra ID

- Azure Monitor
- Azure CLI
- PowerShell
- Cloud management portals
- Infrastructure monitoring tools

Certifications

Certifications related to cloud administration include:

- Microsoft Certified: Azure Administrator Associate
- AWS Certified CloudOps Engineer – Associate
- CompTIA Cloud+
- Google Associate Cloud Engineer

Microsoft's Azure Administrator Associate credential centers on managing Azure identities and governance, storage, compute resources, virtual networking, and the monitoring and upkeep of Azure resources.

Collaboration of System Administration Professionals

Helpdesk Technicians, Network Administrators, Linux System Administrators, and Cloud Administrators don't work in isolation — together they keep the IT environment reliable and secure. Helpdesk Technicians are typically the first point of contact for employees, resolving everyday issues and escalating the harder ones. Network Administrators keep the connections open that let users and systems talk to each other, while Linux System Administrators look after Linux-based servers and the services running on them. Cloud Administrators, meanwhile, manage infrastructure and services hosted in the cloud. When a problem crosses into more than one of these areas, these professionals coordinate to trace the cause and get services back up. Through this shared responsibility and pooled expertise, the four roles keep an organization's users, networks, servers, and cloud resources available, secure, and running smoothly.

Part 7 – Infrastructure Recommendations

The recommendations below take into account Pixelora's current headcount of 20 employees, its software development work, and its need for IT services that are reliable, secure, and able to scale.

Internet Provider

Recommended Provider: PLDT Enterprise Fiber

Pixelora should adopt a business-grade fiber connection from PLDT Enterprise as its main internet service. A fast, dependable connection matters here because the company leans heavily on cloud services, development platforms, online communication, file transfers, video conferencing, and other internet-dependent tools.

A business fiber plan fits better than a residential-grade connection since it's built for business use and leaves room to accommodate the company as it grows.

As reliance on online services deepens, the company may also want to consider a secondary connection from a different provider as a fallback in case the primary line goes down.

Server Specifications

A dedicated server is recommended to run Pixelora's internal services and centralized resources.

Recommended Specifications:

- Processor: Intel Xeon E-2400 series or equivalent server-grade processor
- Memory: 32 GB ECC RAM
- Storage: 2 × 2 TB enterprise-grade SSDs configured in RAID 1
- Network: Dual Gigabit Ethernet ports
- Power Supply: Reliable server-grade power supply
- Operating System: Ubuntu Server 24.04 LTS
- UPS Protection: Dedicated UPS connection

This server will handle centralized file services, internal applications, user management, development-related services, and other company resources. The ECC memory adds a layer of reliability, and RAID 1 provides redundancy by mirroring data across both drives.

The server should also leave room for future memory and storage upgrades as the company's needs grow.

Backup Strategy

Pixelora should follow a 3-2-1 backup strategy for its important data.

In practice, this means keeping:

- 3 copies of important company data
- 2 different types of storage media
- 1 backup copy stored off-site

The primary copy stays on the company server or NAS. A second copy can live on the local NAS or on external backup storage, while a third, encrypted copy should be kept securely off-site or with a trusted cloud backup provider.

Critical files should be backed up automatically every day, with complete backups run on a regular schedule. The two external backup drives listed in the hardware inventory can also be rotated, keeping one safely away from the primary infrastructure at all times.

Backup restoration should be tested from time to time to confirm that files can genuinely be recovered when the need arises.

Security Recommendations

Rather than leaning on one security solution, Pixelora should apply several layers of protection.

Recommended security measures include:

- Configure and maintain the hardware firewall.
- Enable multifactor authentication for important company accounts.
- Apply operating system and software security updates regularly.
- Restrict administrator privileges to authorized IT personnel.
- Separate employee and guest wireless access.
- Control access to company files according to employee roles.
- Encrypt sensitive company information and backup data.
- Disable unused accounts and network services.
- Monitor systems and network activity for unusual behavior.
- Provide employees with basic cybersecurity awareness training.

These measures make it less likely that a single compromised account, device, or application could put the entire company infrastructure at risk.

Antivirus and Endpoint Protection

Recommended Solution: Microsoft Defender Antivirus

Defender Antivirus should stay enabled and be centrally managed across the company's Windows 11 Pro desktops and laptops. Already built into Windows, it guards against malware and other common endpoint threats.

Real-time protection, security updates, and routine scans should remain active at all times, and employees should not be able to turn off endpoint protection without IT Department authorization.

Antivirus software is only one piece of the puzzle — it should work alongside the firewall, software updates, access controls, backups, and employee security awareness as part of the company's overall security strategy.

Password Policy

Pixelora should adopt a company-wide password and authentication policy along these lines:

- Require passwords of at least 14 characters.
- Encourage long and unique passwords or passphrases.
- Block common and easily guessed passwords.
- Do not allow employees to reuse company passwords for personal accounts.
- Require multifactor authentication for important accounts and services.
- Use separate administrator accounts for administrative activities.
- Never share passwords between employees.
- Change passwords immediately when an account is suspected of being compromised.
- Use an approved password manager when employees need to maintain multiple credentials.

Following these practices makes passwords harder to guess and reduces the risk of credential reuse and account compromise, without piling on unnecessary complexity for employees.

Expansion Plan

Pixelora's infrastructure should be built so the company can grow into it rather than needing to replace the whole environment down the line.

The 48-port managed switch chosen for the network already has more capacity than the company currently needs, leaving room to connect additional workstations and network devices later on.

The server should support added memory and storage, and the NAS should offer expandable capacity as well. More wireless access points can be added if the company moves to a larger office or takes on an additional floor.

As headcount grows, Pixelora can also add more servers, segment the network with VLANs, adopt more advanced centralized device management, expand backup capacity, and grow its cloud infrastructure.

Should the company eventually open additional office locations, secure VPN connections or similar business networking solutions can link employees and company resources across sites.

Planning for growth from the outset lets Pixelora scale its infrastructure gradually as business needs increase, rather than replacing major components every time more capacity is required.

Part 8 – Personal Reflection

Working through this Enterprise Infrastructure Planning project showed me just how much groundwork goes into properly deploying an organization's IT infrastructure. I came to realize that setting up a company's technology isn't as simple as buying computers and plugging them into the internet — a System Administrator first has to understand what the organization actually needs, then decide which hardware, software, networking equipment, security measures, and backup solutions fit those operations.

The part I found most demanding was designing the enterprise network topology. I had to work out how the different network components should connect and how devices from each department would communicate across the network. Building the topology also deepened my understanding of what the modem, router, firewall, managed switch, wireless access points, and server each contribute — instead of thinking of them as separate boxes, I started to see how they function together as one network.

I also picked up on how much organizational needs should drive equipment and software choices. Factors like employee count, department, and the nature of the work all shape decisions on the number of computers needed, software licenses, network capacity, storage, and backup requirements. Getting this planning right helps avoid unnecessary spending while still leaving room for future growth.

Planning ahead of deployment matters because mistakes made at this stage can ripple out and affect the reliability, security, and scalability of the whole infrastructure. A clear infrastructure plan makes it much easier to spot requirements, potential security risks, backup needs, and expansion opportunities before any equipment is bought or installed.

This project pushed me to look at IT infrastructure as one connected system rather than a collection of separate devices and applications, and it sharpened my grasp of documentation, network planning, security, hardware and software selection, and scalability. These are skills I plan to keep building on as I take on larger and more complex IT environments in the future.